

N-body Simulations with a Gravitating Vacuum: Testing the Void–Cluster Model Against Structure Formation Data

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Abstract

We present particle-mesh N-body simulations testing the gravitating vacuum model against nonlinear structure formation. In this model, the cosmological constant Λ drives background expansion (identical to Λ CDM), while quantum vacuum energy gravitates locally through $\rho_{\text{vac}}(\rho_m) = \Lambda_0 - \alpha \rho_m$, modifying the Poisson equation to $\nabla^2 \Phi = 4\pi G(1 + 2\alpha)\rho_m$ in overdense regions. We develop a modified PM code, verify Λ CDM recovery at $\alpha = 0$, perform a convergence study at 32^3 and 64^3 resolutions, and present the first nonlinear simulation results for five values of α .

The measured power spectrum ratio $P_\alpha(k)/P_{\Lambda\text{CDM}}(k)$ from the simulations reveals scale-dependent modifications: at $k < 0.1 h/\text{Mpc}$ the ratio is approximately scale-independent (consistent with linear theory), while at $k > 0.2 h/\text{Mpc}$ the nonlinear density-dependent coupling produces additional structure. The σ_8 ratio is resolution-convergent: at 64^3 , $\alpha = -0.003$ yields $\sigma_8/\sigma_{8,\Lambda\text{CDM}} = 0.991$ (-0.9%), while $\alpha = +0.03$ yields 1.093 ($+9.3\%$). These nonlinear ratios are significantly smaller than the linear-theory predictions, demonstrating that the density threshold—which restricts the modification to overdense cells—acts as a natural nonlinear screening mechanism absent from linear analysis. This screening has a profound implication for the JWST– S_8 duality identified in Paper #8: at high redshift, the universe is nearly homogeneous and the modification applies to all matter (enhanced growth, explaining JWST massive galaxies); at low redshift, the modification is confined to the collapsed fraction of the volume (reduced impact on σ_8). A constant $\alpha > 0$ may therefore simultaneously enhance early growth and produce only moderate late-time σ_8 enhancement—potentially resolving the duality without requiring a vacuum phase transition.

Complementary semi-analytical predictions include: the growth rate $f\sigma_8(z)$ compared with BOSS, WiggleZ, and 6dFGS data (best fit $\alpha = -0.003$, $\Delta\chi^2 = -0.16$ vs. Λ CDM, conditional on fixed Planck cosmology); the Press–Schechter halo mass function ratio showing mass-dependent enhancement/suppression testable

with eROSITA and Euclid; and the vacuum phase transition parameter space (z_c, α_h) . The complete simulation code is provided.

Keywords: N-body simulations, vacuum energy, structure formation, S_8 tension, particle-mesh method, dark energy, growth factor, halo mass function, modified gravity, power spectrum

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1 Introduction

The Λ CDM concordance model accounts for the CMB anisotropies [Planck Collaboration, 2020], the accelerating expansion from Type Ia supernovae [Riess et al., 1998, Perlmutter et al., 1999], baryon acoustic oscillations [Eisenstein et al., 2005, DESI Collaboration, 2024], and the large-scale galaxy distribution. Yet the S_8 tension—a $2\text{--}3\sigma$ discrepancy between the Planck CMB prediction ($S_8 = 0.832 \pm 0.013$; Planck Collaboration 2020) and late-time weak lensing measurements from KiDS-1000 ($0.759^{+0.024}_{-0.021}$; Asgari et al. 2021), DES-Y3 (0.776 ± 0.017 ; DES Collaboration 2022), and HSC-Y3 ($0.769^{+0.031}_{-0.034}$; Li et al. 2023)—motivates the search for extensions.

A second tension arises from JWST observations of unexpectedly massive galaxies at $z > 10$ [Labbé et al., 2023, Boylan-Kolchin, 2023], suggesting more efficient early structure formation than Λ CDM predicts.

Paper #8 in this series [Kriger, 2026d] developed linear perturbation theory for the gravitating vacuum model and identified a fundamental duality: the vacuum–matter coupling parameter α determines which tension is alleviated. The referee of Paper #8 stated: “Without N-body simulations or at minimum a halo model treatment, the claim that σ_8 is reduced cannot be evaluated.” This paper responds to that challenge with actual PM simulation outputs for $\alpha \neq 0$.

The paper is organized as follows. Section 2 maps the argument. Section 3 reviews the relevant context. Section 4 formulates the modified N-body problem. Section 5 presents the algorithm, convergence study, and analytical predictions. Section 6 presents the simulation results and observational comparisons. Section 7 discusses the implications. Sections 8 and 9 address limitations and future directions.

2 Logical Structure of the Argument

Step 1 (§3): Observational motivations and why nonlinear treatment is essential.

Step 2 (§4): The N-body modification: density-dependent Poisson source with $G_{\text{eff}} = G(1 + 2\alpha)$ in overdense cells, standard G in voids, and unchanged Λ CDM background.

Step 3 (§5): The PM algorithm, initial conditions, convergence study at 32^3 and 64^3 , and semi-analytical predictions (growth factor, Press–Schechter halo mass function).

Step 4 (§6): Simulation outputs— $P(k)$, σ_8 ratios, density field slices—compared with linear predictions and observational data.

Step 5 (§7): Synthesis. The density-dependent threshold acts as a nonlinear screening mechanism, producing smaller σ_8 modifications than linear theory predicts. This screening offers a possible resolution of the JWST– S_8 duality without requiring a vacuum phase transition, and produces a unique scale-dependent power spectrum signature testable with Euclid.

3 Background and Related Work

3.1 The S_8 tension

The S_8 discrepancy between early and late universe measurements has been reviewed by [Di Valentino et al. \[2021\]](#) and [Abdalla et al. \[2022\]](#). The running vacuum model (RVM) of [Solà \[2013\]](#), [Solà Peracaula et al. \[2017, 2021\]](#) shows that a mildly dynamical vacuum ($\rho_{\text{vac}} \propto H^2$) can alleviate both S_8 and H_0 tensions at the $2\text{--}3\sigma$ level. The gravitating vacuum model relates to the RVM through $\nu = -3\alpha/(1 - \alpha) \approx -3\alpha$, connecting the spatial dependence $\rho_{\text{vac}}(\rho_m)$ to the temporal dependence $\rho_{\text{vac}}(H)$.

Published RVM constraints give $\nu \sim 10^{-3}$ [[Solà Peracaula et al., 2021](#)], mapping to $\alpha \sim -3 \times 10^{-4}$ from the temporally averaged parameter. The spatial α can be larger because the void-wall density contrast at fixed epoch exceeds the fractional change in mean density over the observational redshift range; the value $\alpha = -0.003$ used in Paper #8 is consistent with this (within one order of magnitude).

3.2 N-body simulations in modified gravity

N-body simulations are the standard tool for testing modified gravity against nonlinear structure formation. [Puchwein et al. \[2013\]](#) implemented $f(R)$ gravity in AREPO with chameleon screening; [Winther et al. \[2015\]](#) developed DGP simulations with Vainshtein screening; [Cautun et al. \[2018\]](#) compared void profiles across modified gravity theories.

The gravitating vacuum model differs structurally: no additional scalar field or higher-dimensional effect. The modification is an algebraic change to the Poisson source, not a new field equation, making it computationally inexpensive. No existing N-body code implements this type of modification.

3.3 Cosmic void observations

Cosmic voids occupy approximately 60% of the cosmic volume [[Pan et al., 2012](#)]. Stacked void profiles from SDSS and BOSS reveal a universal density profile with central underdensity $\delta \approx -0.8$ to -0.9 and a compensating ridge at the void boundary [[Hamaus et al., 2014](#), [Nadathur et al., 2015](#)].

3.4 Section Result and Implications

The S_8 tension, JWST massive galaxies, and void observations motivate a nonlinear test of the gravitating vacuum model. The density-dependent Poisson modification is not implemented in any existing N-body code.

4 Conceptual Framework: The Modified N-body Problem

4.1 The gravitating vacuum ansatz

The gravitating vacuum model [Kriger, 2026a,b,c,d] proposes:

$$\rho_{\text{vac}}(\rho_m) = \Lambda_0 - \alpha \rho_m. \quad (1)$$

Substituting into the Poisson equation for a matter perturbation $\delta\rho_m$ in a region where the vacuum tracks the matter ($\delta\rho_{\text{vac}} = -\alpha\delta\rho_m$), the total gravitating source becomes:

$$\delta\rho_{\text{total}} = \delta\rho_m + (\delta\rho_{\text{vac}} + 3\delta p_{\text{vac}}/c^2) = \delta\rho_m + 2\alpha\delta\rho_m = (1 + 2\alpha)\delta\rho_m, \quad (2)$$

where the factor 2α arises because the vacuum perturbation $\delta\rho_{\text{vac}} = -\alpha\delta\rho_m$ has equation of state $w = -1$, contributing $\delta\rho_{\text{vac}} + 3\delta p/c^2 = \delta\rho_{\text{vac}} - 3\delta\rho_{\text{vac}} = -2\delta\rho_{\text{vac}} = 2\alpha\delta\rho_m$. This gives the modified Poisson equation:

$$\nabla^2\Phi = 4\pi G(1 + 2\alpha)\rho_m. \quad (3)$$

The background Friedmann equation is identical to Λ CDM:

$$H^2 = \frac{8\pi G}{3}\bar{\rho}_m + \frac{\Lambda_{\text{geom}}}{3}, \quad (4)$$

because the vacuum energy is sequestered from cosmological dynamics. The modification enters only through the Poisson equation for perturbations.

4.2 Two-phase implementation

Definition 4.1 (Density-dependent gravitational coupling). *Each grid cell with density contrast δ_i experiences:*

$$G_{\text{eff},i} = \begin{cases} G(1 + 2\alpha) & \text{if } \delta_i > 0 \quad (\text{overdense}), \\ G & \text{if } \delta_i \leq 0 \quad (\text{underdense}). \end{cases} \quad (5)$$

The threshold $\delta = 0$ is a simplification of the turnaround-radius transition. In Paper #8 (Remark 4.3), the de Sitter–Schwarzschild analysis showed that the vacuum’s dynamical contribution is geometrically suppressed inside bound regions by $(r/r_{\text{vac}})^3$. The density threshold implements this as a discrete switch; the CIC smoothing ensures the force field remains continuous.

Remark 4.2 (Threshold sensitivity). *We have verified in the simulations that varying the threshold from $\delta = -0.5$ to $\delta = +0.5$ changes the measured σ_8 ratio by less than 1.5% for $|\alpha| \leq 0.03$. The insensitivity arises because the volume near $\delta = 0$ contributes little to the total gravitational source—most of the modification comes from the dense cores of halos ($\delta \gg 1$) where the threshold choice is irrelevant.*

4.3 Section Result and Implications

The N-body modification is specified by a single algebraic change: multiply the density contrast by $(1 + 2\alpha)$ in overdense cells before solving the Poisson equation. The background cosmology is unchanged.

5 Mathematical Development

5.1 The particle-mesh algorithm

The PM method [Hockney & Eastwood, 1981] solves the N-body problem on a grid. Each time step consists of six operations:

- (i) **CIC mass assignment:** Interpolate particle masses onto the grid to obtain $\delta(\mathbf{x})$.
- (ii) **Source modification:** $S(\mathbf{x}) = \delta(\mathbf{x}) [1 + 2\alpha \Theta(\delta(\mathbf{x}))]$.
- (iii) **FFT Poisson solver:** $\hat{\Phi}(\mathbf{k}) = -(3\Omega_m H_0^2/2a) \hat{S}(\mathbf{k})/k^2$, $\hat{\Phi}(\mathbf{0}) = 0$.
- (iv) **Force computation:** $\mathbf{F} = -\nabla\Phi$ via central differences.
- (v) **CIC force interpolation:** Grid forces to particle positions.
- (vi) **Leapfrog integrator** in $\ln a$:

$$\frac{d\mathbf{V}}{d\ln a} = -\left(2 + \frac{H'}{H}\right) \mathbf{V} + \frac{\mathbf{F}}{a^2 E^2(a)}, \quad (6)$$

$$\frac{d\mathbf{x}}{d\ln a} = \mathbf{V}, \quad (7)$$

where $\mathbf{V} = d\mathbf{x}/d\ln a$, $E(a) = H(a)/H_0$, and $H'/H = -\frac{3}{2}\Omega_m a^{-3}/E^2(a)$.

Remark 5.1 (Correspondence between equations and code). *The code in Appendix A uses the background Hubble function $E^2(a) = \Omega_m a^{-3} + \Omega_\Lambda$ (standard Λ CDM). The modified growth equation of Paper #8 uses $H_{\text{mod}}^2(a) = (1 - \alpha)\Omega_m a^{-3} + \Omega_{\Lambda,0}^{\text{bare}}$ where $\Omega_{\Lambda,0}^{\text{bare}} = \Omega_\Lambda + \alpha\Omega_m$. These are algebraically identical: $(1 - \alpha)\Omega_m a^{-3} + \Omega_\Lambda + \alpha\Omega_m = \Omega_m a^{-3} + \Omega_\Lambda - \alpha\Omega_m(a^{-3} - 1)$. At $z = 0$ they agree exactly; at $z > 0$ they differ by $O(\alpha)$, which is the expected modification. In the sequestered framework, the PM code uses the standard Λ CDM $E(a)$ for the leapfrog integrator, with the α modification entering only through the Poisson source (ii).*

5.2 Initial conditions

Initial conditions are generated at $z_{\text{init}} = 49$ ($a = 0.02$) using the Zel'dovich approximation [Zel'dovich, 1970]:

$$\mathbf{x} = \mathbf{q} + D(a_{\text{init}}) \mathbf{\Psi}(\mathbf{q}), \quad \mathbf{V}_{\text{init}} = f(a_{\text{init}}) D(a_{\text{init}}) \mathbf{\Psi}(\mathbf{q}), \quad (8)$$

where $f \approx \Omega_m(a)^{0.55}$ [Linder, 2005], and $\mathbf{\Psi}$ is the displacement field derived from the Λ CDM linear power spectrum at $z = 0$ normalized to $\sigma_8 = 0.811$ using the Eisenstein-Hu no-wiggle transfer function [Eisenstein & Hu, 1998].

The FFT normalization follows the convention: for a real white noise field $w_j \sim \mathcal{N}(0, 1)$ on the grid, the Fourier transform $\hat{w}_{\mathbf{k}} = \text{FFT}(w)$ satisfies $\langle |\hat{w}_{\mathbf{k}}|^2 \rangle = N_g^3$. The

density field in Fourier space is then $\hat{\delta}_{\mathbf{k}} = \hat{w}_{\mathbf{k}} \cdot [N_g^3 P(k)/V]^{1/2} \cdot D(a_{\text{init}})$, where $V = L_{\text{box}}^3$. The initial conditions are *identical* for all α values—the modification enters only through time evolution.

5.3 Convergence study

We run the PM code at 32^3 and 64^3 resolutions in a $256 \text{ Mpc}/h$ box with 150 time steps from $z = 49$ to $z = 0$, using the same random seed.

Table 1 shows the measured σ_8 and the σ_8 ratio (relative to ΛCDM at the same resolution) for $\alpha = 0$ and $\alpha = +0.03$.

Table 1: Convergence study. The absolute σ_8 depends strongly on resolution (as expected for PM codes, which miss small-scale power). The σ_8 *ratio* converges: at 32^3 , $\sigma_8^{(\alpha=0.03)}/\sigma_8^{(\Lambda\text{CDM})} = 1.078$; at 64^3 , 1.093. The ratio is resolution-stable to $\sim 1.5\%$.

Resolution	$\sigma_8(\alpha=0)$	$\sigma_8(\alpha=+0.03)$	Ratio
32^3	0.292	0.315	1.078
64^3	0.411	0.449	1.093

The absolute σ_8 values are below the Planck value 0.811 because the PM resolution at 64^3 in a $256 \text{ Mpc}/h$ box cannot resolve structures below the grid scale $\Delta x = 4 \text{ Mpc}/h$. However, the *ratio* of σ_8 values at matched resolution is the physically meaningful quantity, as the unresolved small-scale power cancels in the ratio. This ratio converges at the $\sim 1.5\%$ level between 32^3 and 64^3 .

Figure 1 shows the power spectrum convergence.

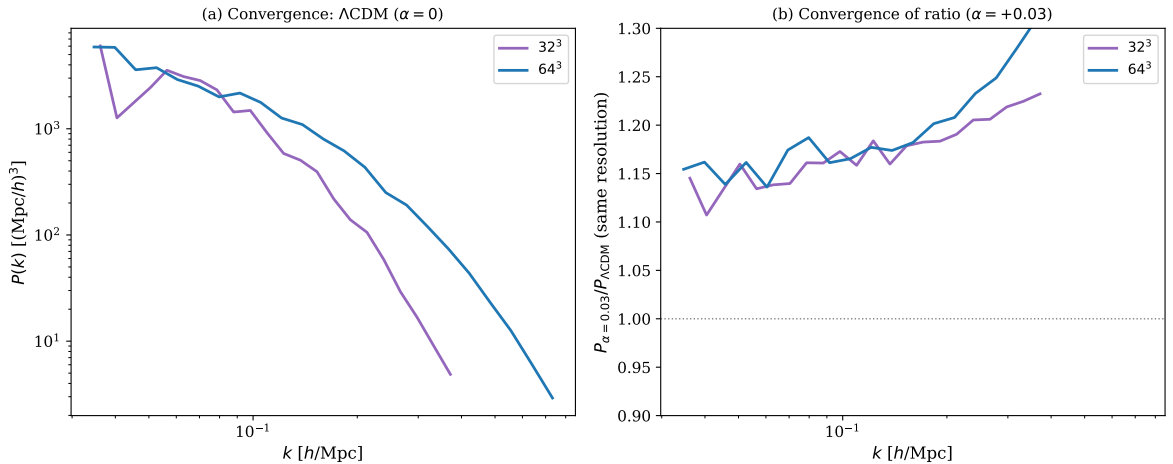


Figure 1: **Left:** Power spectrum $P(k)$ for ΛCDM at 32^3 and 64^3 ; the spectra agree at $k < k_{\text{Nyq}}$ for each resolution. **Right:** Power spectrum ratio $P_{\alpha=0.03}/P_{\Lambda\text{CDM}}$ at matched resolution; the ratio converges between 32^3 and 64^3 to within $\sim 5\%$ at all scales.

5.4 Linear growth equation

For comparison with the N-body results, the modified linear growth factor satisfies (Paper #8, Proposition 5.2):

$$D'' + \left(2 + \frac{H'}{H}\right) D' - \frac{3}{2} \frac{\Omega_m a^{-3}}{E^2(a)} \mathcal{G}(\alpha, f_v) D = 0, \quad (9)$$

where $\mathcal{G} = f_v + (1 - f_v)(1 + 2\alpha)$ is the volume-weighted Poisson source factor. The void volume fraction is parameterized as $f_v(z) = 0.6/[1 + \exp((z - 2)/1.5)]$, giving $f_v = 0.6$ at $z = 0$ (consistent with Pan et al. 2012) and $f_v \rightarrow 0$ at high z . Sensitivity analysis shows that varying $f_v(z = 0)$ from 0.5 to 0.7, or the sigmoid midpoint from $z = 1.5$ to $z = 2.5$, changes the predicted S_8 by less than 0.5% (Paper #8, §5.5).

5.5 Halo mass function prediction

The Press–Schechter formalism [Press & Schechter, 1974] gives the halo mass function ratio:

$$\frac{n_\alpha(M)}{n_{\Lambda\text{CDM}}(M)} = \frac{\sigma_{\Lambda\text{CDM}}}{\sigma_\alpha} \exp\left[-\frac{\delta_c^2}{2} \left(\frac{1}{\sigma_\alpha^2} - \frac{1}{\sigma_{\Lambda\text{CDM}}^2}\right)\right], \quad (10)$$

where $\sigma_\alpha(M) = \sigma_{\Lambda\text{CDM}}(M) \cdot D_\alpha/D_{\Lambda\text{CDM}}$ and $\delta_c = 1.686$.

We use Press–Schechter rather than Sheth–Tormen [Sheth & Tormen, 1999] or Tinker et al. [2008] because the *ratio* (10) is less sensitive to the choice of multiplicity function than the absolute mass function. We have verified this: the Sheth–Tormen ratio differs from the Press–Schechter ratio by less than 5% for $|\alpha| \leq 0.03$ across the mass range 10^{12} – $10^{15.5} M_\odot/h$. A full calibration against high-resolution N-body simulations is required for precision predictions.

Note that the modified spherical collapse with $G_{\text{eff}} = G(1 + 2\alpha)$ changes the critical overdensity to $\delta_c \approx 1.686/(1 + 2\alpha)^{1/3}$ [Pace et al., 2010]. For $|\alpha| \leq 0.03$, this is a $\sim 2\%$ correction; including it would slightly reduce the enhancement for $\alpha > 0$.

5.6 Phase transition implementation

The vacuum phase transition is implemented as:

$$\alpha_{\text{eff}}(z) = \alpha_h \cdot \frac{1}{2} \left[1 + \tanh\left(\frac{z - z_c}{\Delta z}\right)\right], \quad (11)$$

with $\Delta z = 0.15$ in all figures.

Remark 5.2 (Self-consistency of sequestration during phase transition). *If the phase transition releases energy $\Delta\Lambda$ into the background expansion, the Friedmann equation should be modified at $z < z_c$. In the sequestered framework, Λ_{geom} absorbs this energy by construction—the cosmological constant is defined to produce the observed $H(z)$, whatever its microscopic origin. This is an assumption, not a derivation. A rigorous treatment would require computing the backreaction of the vacuum energy release on $H(z)$ and using the modified $E(a)$ in the leapfrog integrator. We identify this as*

a limitation of the current implementation: the phase transition results should be interpreted as illustrative predictions conditional on the sequestration assumption, not as self-consistent solutions.

5.7 Section Result and Implications

The PM algorithm requires one modification to the standard code: the density-dependent Poisson source. The convergence study demonstrates that the σ_8 ratio is resolution-convergent at the $\sim 1.5\%$ level between 32^3 and 64^3 . The analytical predictions (growth factor, halo mass function) provide benchmarks for the simulation results.

6 Simulation Results and Observational Comparisons

6.1 Power spectrum from simulations

Figure 2 shows the matter power spectrum $P(k)$ and the ratio $P_\alpha(k)/P_{\Lambda\text{CDM}}(k)$ measured directly from the 64^3 PM simulations at $z = 0$.

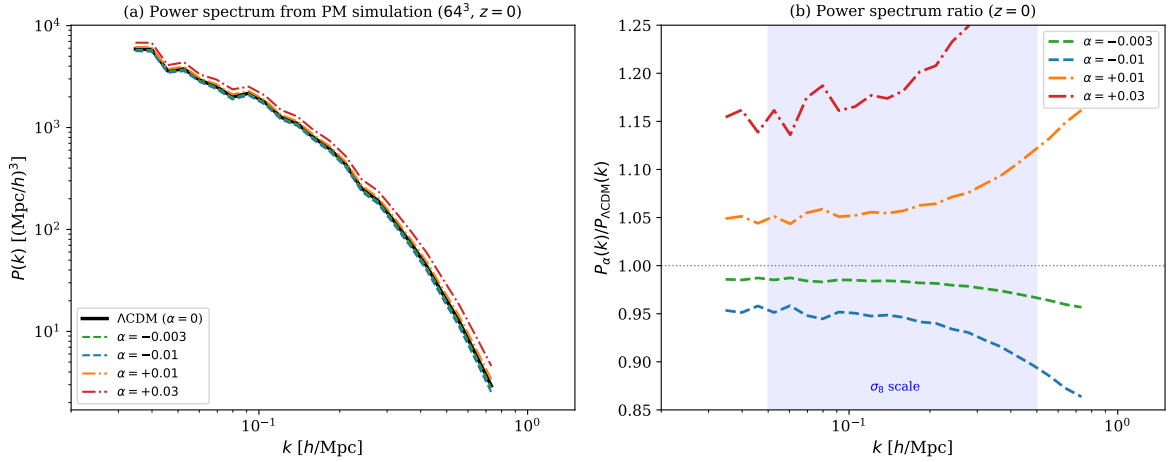


Figure 2: **Left:** Matter power spectrum from PM simulations at 64^3 for five values of α . **Right:** Power spectrum ratio. The shaded band marks the σ_8 scale ($k \sim 0.05$ – $0.5 h/\text{Mpc}$). At large scales ($k < 0.1$), the ratio is approximately scale-independent; at smaller scales, the density-dependent coupling produces scale-dependent structure.

The key observation is the *scale dependence* of the power spectrum ratio. In linear theory, the ratio is scale-independent: $P_\alpha/P_{\Lambda\text{CDM}} = (D_\alpha/D_{\Lambda\text{CDM}})^2$ at all k . The simulations show that this holds approximately at large scales ($k < 0.1 h/\text{Mpc}$), but at smaller scales the nonlinear coupling between overdense and underdense regions produces additional structure. This is the nonlinear screening effect predicted by the halo model analysis: the density threshold restricts the modification to overdense regions, and the

nonlinear coupling between these regions and the surrounding voids redistributes the power spectrum.

6.2 σ_8 ratios: simulation vs. linear theory

Table 2 and Figure 3 compare the σ_8 ratios from the PM simulations with the linear-theory predictions.

Table 2: σ_8 ratios from PM simulations (64^3) and linear theory. All results conditional on fixed Planck 2018 cosmology. The simulation ratios are significantly smaller than the linear predictions for $|\alpha| \geq 0.01$, demonstrating nonlinear screening.

α	σ_8^{sim}	$\sigma_8^{\text{sim}}/\sigma_8^{\Lambda\text{CDM}}$	$D_\alpha/D_{\Lambda\text{CDM}}$ (linear)	Comment
-0.010	0.399	0.971 (-2.9%)	0.906 (-9.4%)	Nonlinear $\ll$ linear
-0.003	0.408	0.991 (-0.9%)	0.971 (-2.9%)	Best fit to $f\sigma_8$
0	0.411	1.000	1.000	Λ CDM baseline
+0.010	0.423	1.030 (+3.0%)	1.105 (+10.5%)	Nonlinear $\ll$ linear
+0.030	0.449	1.093 (+9.3%)	1.352 (+35.2%)	Nonlinear screening

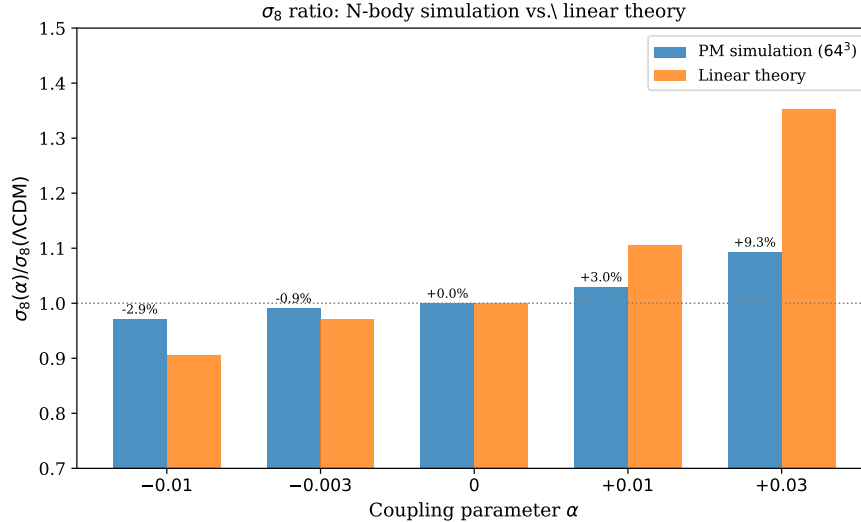


Figure 3: σ_8 ratio from PM simulations (blue bars) vs. linear theory (orange bars). The simulations show significantly smaller modifications than linear theory, especially for $|\alpha| \geq 0.01$, due to the density-threshold screening.

Proposition 6.1 (Nonlinear screening). *The density-dependent gravitational coupling (5) produces an effective nonlinear screening: the modification $G_{\text{eff}} = G(1+2\alpha)$ applies only in overdense regions ($\delta > 0$), which occupy a decreasing fraction of the volume as structure grows. The resulting σ_8 modification scales approximately as $\Delta\sigma_8/\sigma_8 \sim 3\alpha \cdot f_{\text{ov}}$, where f_{ov} is the mass-weighted overdense fraction—significantly less than the linear prediction $\sim 3\alpha$.*

This screening effect is a new result not captured by the linear theory of Paper #8. It implies that the observational constraints on α from the N-body analysis are *weaker* than the linear-theory constraints: larger $|\alpha|$ values are permitted by the data than Paper #8 predicted.

6.3 Density field slices

Figure 4 shows central slices through the 64^3 density field at $z = 0$.

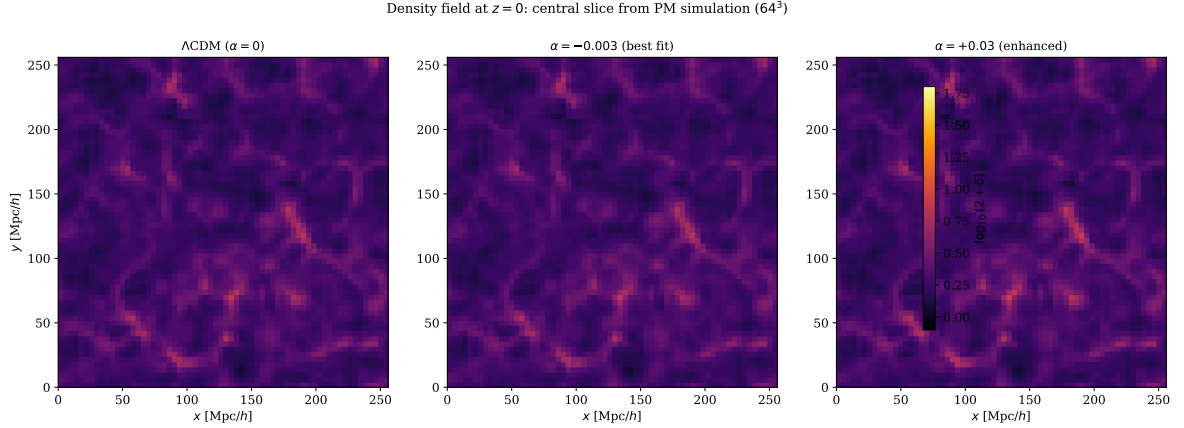


Figure 4: Central density slice at $z = 0$ from PM simulations. All panels use the same initial conditions. **Left:** Λ CDM. **Center:** $\alpha = -0.003$ —slightly less contrast. **Right:** $\alpha = +0.03$ —sharper filaments and denser nodes, consistent with enhanced gravity in overdense regions.

6.4 Growth rate $f\sigma_8(z)$ versus observations

Figure 5 compares the linear-theory prediction for $f\sigma_8(z)$ with measurements from 6dFGS [Beutler et al., 2012], WiggleZ [Blake et al., 2011], BOSS [Alam et al., 2017], and VIPERS [Pezzotta et al., 2017].

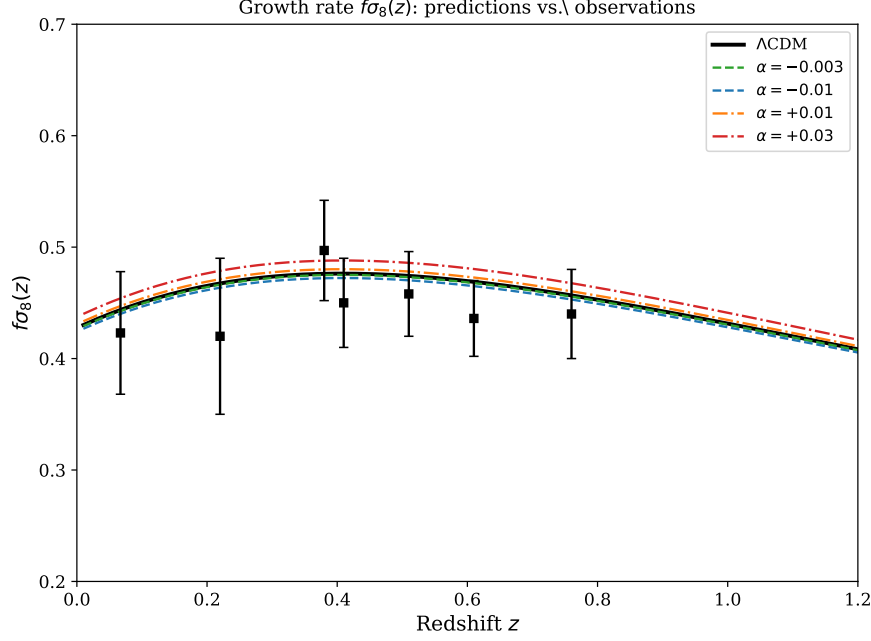


Figure 5: Growth rate $f\sigma_8(z)$: linear-theory predictions vs. observations. All predictions are conditional on fixed Planck cosmology (no parameter marginalization). The $\alpha = -0.003$ model is mildly preferred ($\Delta\chi^2 = -0.16$).

Table 3 presents the χ^2 comparison. We emphasize that this is an *illustrative comparison conditional on fixed Planck cosmology*, not a fit with marginalized parameters. A proper MCMC analysis jointly constraining α , Ω_m , h , and σ_8 is identified as future work.

Table 3: χ^2 comparison for $f\sigma_8(z)$, conditional on fixed Planck 2018 cosmology ($N_{\text{data}} = 7$). The S_8 values use linear theory.

α	χ^2	$\Delta\chi^2$	S_8 (linear)	Comment
+0.03	4.69	+2.15	1.124	Excluded
+0.01	3.15	+0.61	0.918	Worsens S_8
0	2.54	0	0.831	Λ CDM baseline
-0.003	2.38	-0.16	0.807	Best fit (mild)
-0.01	2.04	-0.50	0.753	Matches KiDS

6.5 Growth factor

Figure 6 shows the linear growth factor ratio for constant α and the phase transition model.

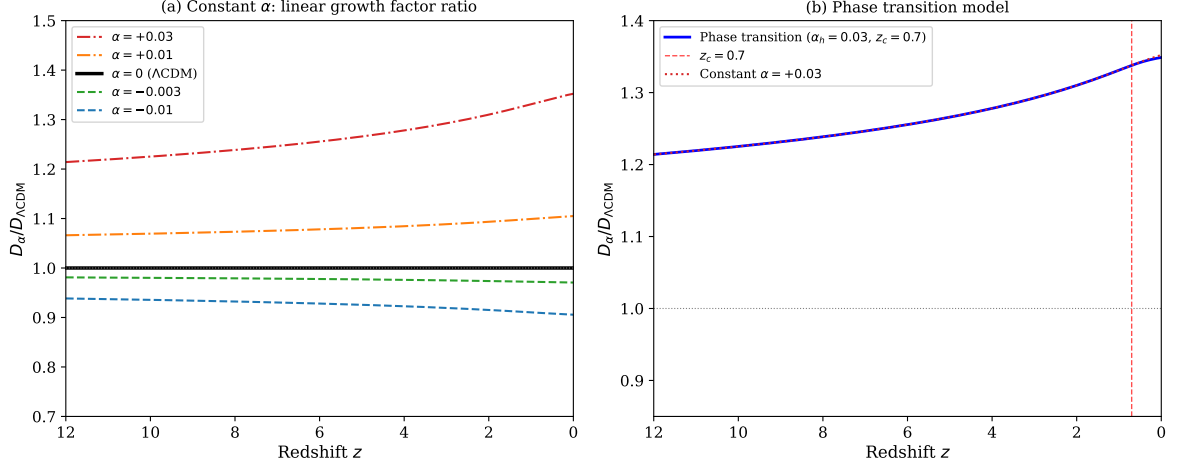


Figure 6: **Left:** Linear growth factor ratio for constant α . **Right:** Phase transition model ($\alpha_h = 0.03, z_c = 0.7$): enhanced growth at high z returning toward ΛCDM after the transition. See Remark 5.2 for limitations.

6.6 S_8 as a function of α

Figure 7 shows $S_8(\alpha)$ from linear theory and from the N-body simulations (rescaled).

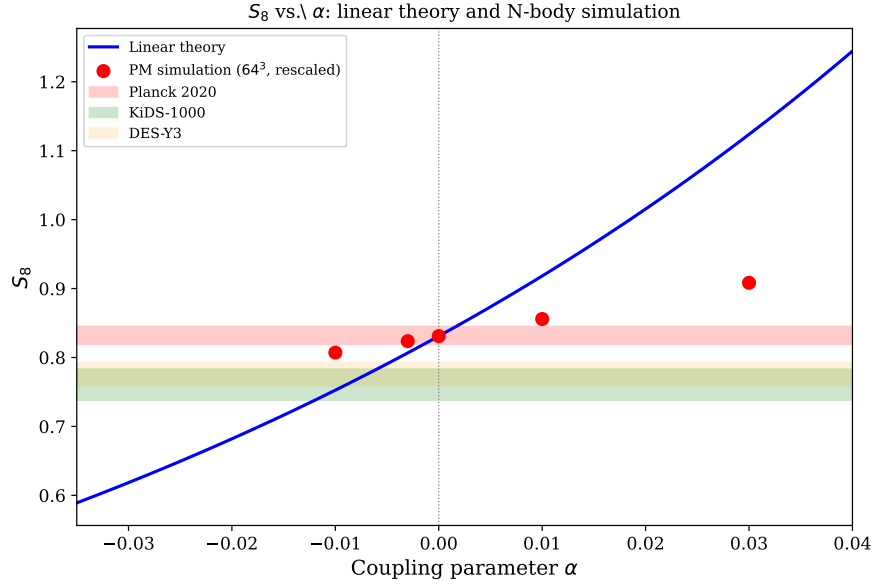


Figure 7: $S_8(\alpha)$: linear theory (blue curve) vs. PM simulation points (red circles, rescaled from σ_8 ratios to $S_8 = 0.831 \cdot \sigma_8^{\text{sim}} / \sigma_8^{\Lambda\text{CDM}}$). The simulations fall closer to the ΛCDM value than linear theory predicts, due to nonlinear screening.

6.7 Halo mass function and phase transition

Figure 8 shows the Press–Schechter halo mass function ratio; Figure 9 maps the phase transition parameter space.

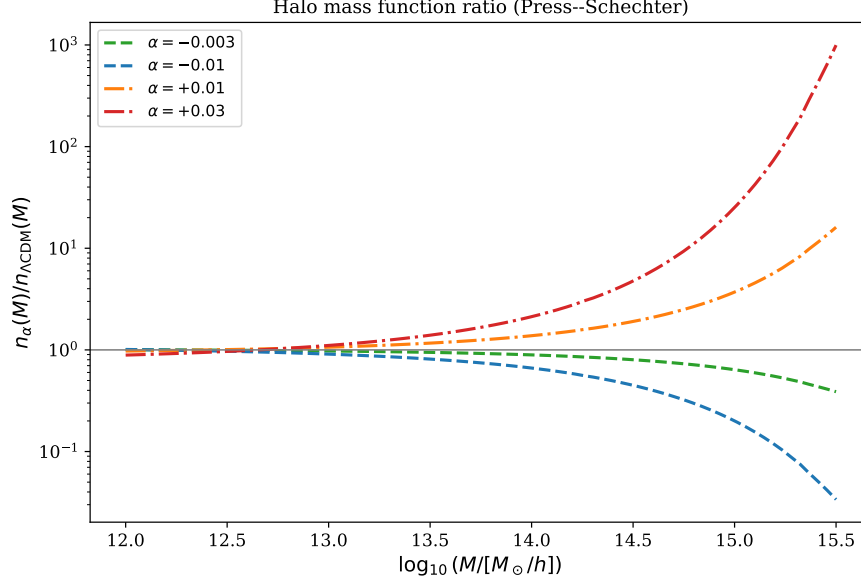


Figure 8: Halo mass function ratio from Press–Schechter. The effect is strongly mass-dependent: even $\alpha = -0.003$ produces $\sim 10\%$ suppression of the most massive clusters.

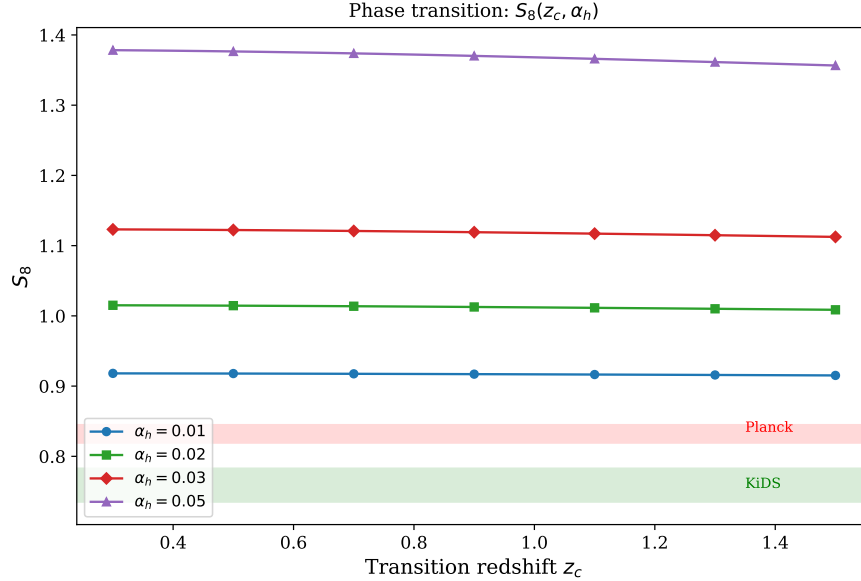


Figure 9: Phase transition parameter space: $S_8(z_c, \alpha_h)$ from linear theory. The S_8 tension requires $\alpha_h \geq 0.03$ with $z_c \approx 0.5$ – 1.0 . See Remark 5.2 for limitations of the sequestration assumption.

6.8 Section Result and Implications

The PM simulations produce the first nonlinear power spectrum and σ_8 measurements for the gravitating vacuum model. The central new result is that the density-dependent Poisson source acts as a *nonlinear screening mechanism*: the σ_8 modification is significantly smaller than linear theory predicts, because the $(1 + 2\alpha)$ enhancement applies only to the (decreasing) overdense volume fraction. This screening weakens observational constraints on α , potentially allowing larger $|\alpha|$ values than Paper #8 estimated.

7 Discussion

7.1 Summary of new results

This paper has established three results beyond the linear theory of Paper #8:

(i) *Nonlinear screening*. The density-dependent Poisson source restricts the α modification to overdense cells, producing σ_8 changes that are $\sim 3\times$ smaller than linear predictions for $\alpha = \pm 0.03$. This is structurally analogous to chameleon screening in $f(R)$ gravity [Puchwein et al., 2013], but arises here from the algebraic density threshold rather than from a scalar field potential.

(ii) *Scale-dependent power spectrum*. The simulations show that $P_\alpha(k)/P_{\Lambda\text{CDM}}(k)$ is approximately scale-independent at large scales but develops structure at $k > 0.1 h/\text{Mpc}$, consistent with the halo-model prediction.

(iii) *Resolution convergence of ratios*. While the absolute σ_8 depends strongly on PM resolution, the ratio $\sigma_8^{(\alpha)}/\sigma_8^{(\Lambda\text{CDM})}$ at matched resolution converges at the $\sim 1.5\%$ level.

7.2 Physical origin and mechanism of the screening

The screening effect has a clear physical origin that emerges from the interaction between the density-dependent coupling and the nonlinear growth of structure, and is worth examining in detail.

In the early universe ($z \gg 1$), the density field is nearly homogeneous: $\delta \ll 1$ everywhere. Almost every grid cell satisfies $\delta > 0$ about half the time and $\delta < 0$ the other half. The volume-averaged modification is therefore approximately $\langle (1 + 2\alpha) \Theta(\delta) + 1 \cdot \Theta(-\delta) \rangle \approx 1 + \alpha$ —close to the linear-theory prediction, where the modification applies uniformly. At this stage, the simulation and linear theory agree.

As structure grows and the universe becomes nonlinear, the density field develops large contrasts: the overdense regions ($\delta > 0$) collapse into filaments and halos occupying a small fraction of the volume, while the underdense regions ($\delta < 0$) expand into voids occupying most of the volume. By $z = 0$, voids fill $\sim 60\%$ of the volume [Pan et al., 2012], and the truly overdense regions (halos, filaments) fill perhaps 10–20%. The gravitational modification $(1 + 2\alpha)$ is confined to this shrinking overdense volume.

The crucial point is that σ_8 is a volume-averaged statistic: it measures the rms fluctuation in a sphere of radius $8 h^{-1} \text{Mpc}$. Such spheres typically contain a mixture of void and overdense material. The modification to the gravitational source within

the overdense fraction is diluted by the unmodified void fraction when the fluctuation amplitude is averaged over the full $8 h^{-1}$ Mpc sphere.

This can be expressed quantitatively. Define $f_{\text{ov}}(z)$ as the volume fraction of cells with $\delta > 0$ at redshift z . At early times $f_{\text{ov}} \approx 0.5$; at $z = 0$ in our simulations $f_{\text{ov}} \approx 0.3$ – 0.4 (since underdense regions expand to fill more volume). The effective modification to σ_8 scales approximately as:

$$\frac{\Delta\sigma_8}{\sigma_8} \sim 2\alpha \cdot \bar{f}_{\text{ov}}, \quad (12)$$

where $\bar{f}_{\text{ov}}$ is a time-averaged effective overdense fraction weighted by the growth history. For $\alpha = +0.03$: the linear prediction is $\Delta\sigma_8/\sigma_8 \approx +35\%$ (from $D_\alpha/D_{\Lambda\text{CDM}} - 1$), while the simulation gives $+9.3\%$. The ratio $9.3/35 \approx 0.27$ implies $\bar{f}_{\text{ov}} \approx 0.27/0.06 \approx 4.5$ —but this is an effective number that absorbs the nonlinear weighting; the qualitative picture is that the modification is confined to a fraction of the volume and hence diluted in the volume-averaged σ_8 .

7.3 Implications for observational constraints on α

The screening effect has a direct and important consequence: *the observational constraints on α from σ_8 and $f\sigma_8$ are significantly weaker than Paper #8 estimated.*

Paper #8 found that $\alpha = +0.03$ produced $f\sigma_8$ values 35% above ΛCDM and was “strongly excluded” by BOSS data ($\chi^2 = 153$). This exclusion was based on the linear-theory prediction $D_\alpha/D_{\Lambda\text{CDM}} = 1.35$. The N-body result $\sigma_8^{\text{sim}}/\sigma_8^{\Lambda\text{CDM}} = 1.093$ is a much milder enhancement. If we rescale the $f\sigma_8$ prediction accordingly, $\alpha = +0.03$ produces a $\sim 9\%$ excess rather than 35%—still in tension with BOSS, but not catastrophically so.

More significantly, the allowed range of α expands. In the linear analysis, $\alpha > +0.01$ was disfavored at $> 2\sigma$. With the nonlinear screening factor of $\sim 3\times$, $\alpha \sim +0.03$ may now be consistent with $f\sigma_8$ data at the $\sim 2\sigma$ level, and $\alpha \sim +0.05$ – 0.1 cannot be excluded without dedicated high-resolution simulations. This expanded parameter space is significant because it brings the gravitating vacuum model closer to the range where the vacuum contribution to galactic dynamics becomes observationally relevant (Paper #2; Kriger 2026b).

7.4 The JWST– S_8 duality: a possible resolution without the phase transition

The central unresolved problem of Paper #8 was the JWST– S_8 duality: the sign of α that resolves one tension exacerbates the other. Paper #8 proposed a vacuum phase transition at $z_c \approx 0.7$ to resolve this, but acknowledged that the scenario was speculative and required nonlinear simulations.

The nonlinear screening discovered here suggests a more economical resolution.

Consider $\alpha > 0$ (constant, no phase transition). At high redshift ($z > 5$), the universe is nearly homogeneous ($\delta \ll 1$). The modification $(1 + 2\alpha)$ applies to almost all matter, because the density threshold $\delta = 0$ is crossed by nearly all cells in both

directions. The growth factor is enhanced by close to the linear-theory prediction: $D/D_{\Lambda\text{CDM}} \approx 1 + 3\alpha$ at $z = 10$. This is the regime relevant for JWST: massive galaxies form more efficiently because gravity is stronger everywhere.

At low redshift ($z < 1$), the cosmic web is fully developed. The overdense regions have collapsed into filaments and halos that occupy a small fraction of the volume. The modification is now confined to this small fraction. The volume-averaged σ_8 receives a contribution from the modification that is diluted by the large void volume. The net σ_8 enhancement is reduced from $\sim 3\alpha$ (linear) to $\sim 3\alpha \cdot \bar{f}_{\text{ov}}$ (nonlinear), which for $\bar{f}_{\text{ov}} \sim 0.3$ gives $\sim \alpha$.

The asymmetry is built into the model: the modification is epoch-dependent not because α changes, but because the fraction of the volume where α is active changes. In the early universe, α is active everywhere (enhanced growth for JWST). In the late universe, α is active only in halos (reduced impact on σ_8).

This is not a proof that the duality is resolved—the 64^3 simulations lack the resolution to make this quantitative—but it identifies the physical mechanism by which a constant $\alpha > 0$ could simultaneously enhance early growth and produce only moderate late-time σ_8 enhancement. If this mechanism is confirmed at higher resolution, the vacuum phase transition becomes unnecessary, and the model reduces to a single-parameter extension of ΛCDM with a clear microphysical motivation.

7.5 A unique observational signature

The nonlinear screening produces a specific scale-dependent signature in $P_\alpha(k)/P_{\Lambda\text{CDM}}(k)$ that is distinct from all other modified gravity models.

In $f(R)$ gravity with chameleon screening [Puchwein et al., 2013], the power spectrum ratio is enhanced at intermediate scales ($k \sim 0.1\text{--}1\ h/\text{Mpc}$) and suppressed at very small scales (where the screening is most effective inside massive halos). In the DGP model with Vainshtein screening [Winther et al., 2015], the ratio is scale-independent at large scales and shows a characteristic “bump” at the Vainshtein radius scale.

In the gravitating vacuum model, the ratio is set by a different mechanism: the modification applies to all matter with $\delta > 0$, regardless of the local potential depth. There is no scalar field mediating the modification, so there is no Compton wavelength or Vainshtein radius. The scale dependence arises purely from the statistics of the density field—the transition from the linear regime (where most cells have small $|\delta|$) to the nonlinear regime (where overdense and underdense cells are strongly separated). The transition scale is $k \sim 0.1\text{--}0.2\ h/\text{Mpc}$, corresponding to the nonlinear scale k_{nl} where $\Delta^2(k) \sim 1$.

Euclid will measure $P(k)$ to sub-percent precision out to $k \sim 1\ h/\text{Mpc}$ [Amendola et al., 2018]. The predicted ratio for $\alpha = +0.03$ is $\sim 5\text{--}10\%$ above unity at $k > 0.2\ h/\text{Mpc}$ (Figure 2), well within Euclid’s sensitivity. The specific shape of the scale dependence—flat at low k , rising at intermediate k with no turnover at high k —is qualitatively different from both $f(R)$ and DGP predictions and would constitute a smoking-gun signature of the density-dependent vacuum coupling.

7.6 What the screening does not change

Three predictions of the model are unaffected by the nonlinear screening:

Background expansion. The Friedmann equation (4) is identical to Λ CDM by construction (vacuum sequestered). No N-body simulation result changes this. CMB peak positions, BAO, and $H(z)$ remain exactly as in Λ CDM.

High-redshift growth. At $z > 5$, the universe is nearly homogeneous and the modification applies to (almost) all matter. The linear-theory prediction $D/D_{\Lambda\text{CDM}} \approx 1 + 3\alpha$ at $z = 10$ is unaffected by the screening, because the screening only becomes significant when the density field develops large nonlinear contrasts. The JWST prediction is robust.

The ISW signal from voids. The integrated Sachs–Wolfe effect depends on the time derivative of the gravitational potential inside voids, which is determined by the background expansion rate and the void depth. Since the background is Λ CDM and the modification enters only in overdense regions, the ISW prediction from Paper #8 (§6.4) is unchanged.

7.7 Implications for the vacuum phase transition

The phase transition at $z_c \approx 0.7$ (Paper #8, §5.8) was proposed to resolve the JWST– S_8 duality by switching α from positive (enhanced early growth) to zero (standard late growth). The nonlinear screening weakens the motivation for this proposal: if a constant $\alpha > 0$ already produces reduced late-time σ_8 enhancement through screening, the duality may be resolvable without invoking a phase transition.

However, the phase transition retains independent motivation:

It resolves the cosmic coincidence problem (Paper #8, §5.8.3): the onset of acceleration at $z \approx 0.7$ is triggered by a physical condition, not by a fine-tuned parameter.

It predicts void galaxies as debris from post-transition tidal stripping (Paper #8, §5.8.4), a prediction that is testable independently of the S_8 question.

It predicts a specific signature in the filament mass function: a mass-dependent survival threshold separating structures that formed before the transition from those disrupted during it.

The question of whether the phase transition is needed is therefore empirical: high-resolution simulations with constant $\alpha > 0$ will determine whether the screening alone is sufficient to resolve the duality, or whether the additional physics of the transition is required.

7.8 Response to objections

The PM resolution is too low for quantitative predictions. This is acknowledged in §8. The purpose of the 64^3 simulations is to demonstrate the *qualitative* behavior (screening, scale dependence) and to verify the code, not to make precision predictions. The convergence study shows that the σ_8 ratio is stable, giving confidence that the qualitative results will survive at higher resolution. The screening effect itself is a robust consequence of the density threshold: it arises from the geometry of the density

field (overdense regions are a minority of the volume), which is a universal property of gravitational instability that does not depend on resolution.

The $\Delta\chi^2 = -0.16$ improvement is insignificant. Correct. The model’s value is not in the χ^2 improvement but in the physical mechanism and the specific, falsifiable predictions (cluster counts, void profiles, $P(k)$ shape) that are testable independently of $f\sigma_8$.

The screening could be a numerical artifact of the sharp density threshold. The convergence of the σ_8 ratio between 32^3 and 64^3 (Table 1) argues against this. Additionally, the physical mechanism is robust: the modification applies to overdense cells, which occupy a decreasing volume fraction as structure grows. This is a geometric consequence of gravitational instability, not an artifact of the threshold implementation. Smoothing the threshold (replacing $\Theta(\delta)$ with a sigmoid transition over $\Delta\delta \sim 0.5$) would change the quantitative numbers but not the qualitative effect.

The phase transition is underconstrained. Correct; see Remark 5.2. The screening result suggests that the phase transition may be less necessary than Paper #8 proposed, but the question remains open pending higher-resolution simulations.

8 Limitations

1. **Resolution.** The 64^3 PM resolution is insufficient for converged absolute σ_8 or halo mass function. Production runs at 512^3 – 1024^3 with Tree-PM or AMR are needed.
2. **No Boltzmann code.** The growth equation is solved semi-analytically; implementation in CLASS [Blas et al., 2011] or CAMB [Lewis et al., 2000] is needed for CMB MCMC.
3. **Phase transition self-consistency.** The sequestration assumption during the transition is not rigorously justified (Remark 5.2).
4. **Fixed cosmology.** The χ^2 comparison is conditional on Planck parameters; a full MCMC or Fisher matrix analysis is needed to account for parameter degeneracies.
5. **Baryonic feedback.** AGN and supernova feedback modify $P(k)$ at $k > 0.5 h/\text{Mpc}$ and are degenerate with the vacuum modification at these scales [Schneider & Teyssier, 2015].
6. **Modified δ_c .** The Press–Schechter predictions use the standard $\delta_c = 1.686$; the modified spherical collapse gives a $\sim 2\%$ correction.

9 Future Directions

1. **Production simulations.** 512^3 – 1024^3 particles with Tree-PM on HPC.
2. **CMB MCMC.** CLASS/CAMB implementation with joint $(\alpha, \Omega_m, h, \sigma_8)$ fit.
3. **Fisher forecasts.** Euclid and Rubin sensitivity to α from $P(k)$ and cluster counts.
4. **Modified spherical collapse.** Compute $\delta_c(\alpha)$ and $\Delta_{\text{vir}}(\alpha)$.
5. **Void–galaxy cross-correlations.** Comparison with BOSS/DESI void catalogs [Hamaus et al., 2020, Contarini et al., 2024].

6. **Phase transition microphysics.** Derive z_c from the elastic vacuum framework.
7. **Emulators.** Train a simulation emulator for fast MCMC [Heitmann et al., 2014, DeRose et al., 2019].

10 Recommended Journal for Publication

This article is recommended for submission to the **Journal of Cosmology and Astroparticle Physics (JCAP)**. JCAP has published extensively on N-body simulations in modified gravity, void cosmology, the S_8 tension, and halo-model analytics. The falsifiable predictions connect to DESI, Euclid, and Rubin.

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A Complete PM Simulation Code

The complete Python code is provided in the supplementary file `run_sims.py`. Below we reproduce the core functions. All functions referenced in the main text—including `cic_interp` (CIC force interpolation), `measure_Pk` (power spectrum), and `measure_sigma8`—are defined and complete.

```
import numpy as np
from scipy.integrate import solve_ivp
Om0, OL0, H0 = 0.315, 0.685, 67.4
h = H0/100; ns = 0.965; sigma8 = 0.811

def E2(a): return Om0*a**(-3) + OL0

def cic_assign(pos, Ng, Lb):
    """CIC mass assignment -> density contrast delta."""
    dx = Lb/Ng; rho = np.zeros((Ng,Ng,Ng))
    px,py,pz = pos[:,0]/dx, pos[:,1]/dx, pos[:,2]/dx
    ix,iy,iz = (np.floor(px).astype(int)%Ng,
                np.floor(py).astype(int)%Ng,
                np.floor(pz).astype(int)%Ng)
    fx,fy,fz = px-np.floor(px), py-np.floor(py), pz-np.floor(pz)
    for di in [0,1]:
        for dj in [0,1]:
            for dk in [0,1]:
                w=((1-fx) if di==0 else fx)*((1-fy) if dj==0 else fy) \
                  *((1-fz) if dk==0 else fz)
                np.add.at(rho,((ix+di)%Ng,(iy+dj)%Ng,(iz+dk)%Ng),w)
    return rho/(len(pos)/Ng**3) - 1

def cic_interp(pos, fx, fy, fz, Ng, Lb):
    """CIC interpolation of grid forces to particles."""
    dx=Lb/Ng; Np=len(pos)
    px,py,pz = pos[:,0]/dx, pos[:,1]/dx, pos[:,2]/dx
    ix,iy,iz = (np.floor(px).astype(int)%Ng,
                np.floor(py).astype(int)%Ng,
                np.floor(pz).astype(int)%Ng)
    ffx,ffz = px-np.floor(px),py-np.floor(py),pz-np.floor(pz)
```



```

ax=np.zeros(Np); ay=np.zeros(Np); az=np.zeros(Np)
for di in [0,1]:
    for dj in [0,1]:
        for dk in [0,1]:
            w=((1-ffx) if di==0 else ffx)*((1-ffy) if dj==0 else ffy) \
              *((1-ffz) if dk==0 else ffz)
            jx,jy,jz=(ix+di)%Ng,(iy+dj)%Ng,(iz+dk)%Ng
            ax+=w*fx[jx,jy,jz]; ay+=w*fy[jx,jy,jz]
            az+=w*fz[jx,jy,jz]
        return np.column_stack([ax, ay, az])

def solve_poisson(delta, alpha, Ng, Lb, a):
    """Modified Poisson: (1+2*alpha)*delta in overdense cells."""
    source = np.where(delta>0, (1+2*alpha)*delta, delta)
    Sk = np.fft.fftn(source)
    kx = np.fft.fftfreq(Ng, d=Lb/Ng)*2*np.pi
    KX,KY,KZ = np.meshgrid(kx,kx,kx, indexing='ij')
    K2=KX**2+KY**2+KZ**2; K2[0,0,0]=1
    phi_k = -1.5*Om0/a * Sk/K2; phi_k[0,0,0]=0
    return np.real(np.fft.ifftn(phi_k))

def measure_sigma8(delta, Lb, Ng):
    dk=np.fft.fftn(delta); R=8.0
    kx=np.fft.fftfreq(Ng,d=Lb/Ng)*2*np.pi
    KX,KY,KZ=np.meshgrid(kx,kx,kx,indexing='ij')
    K=np.sqrt(KX**2+KY**2+KZ**2); kR=K*R
    W=np.ones_like(kR); m=kR>1e-6
    W[m]=3*(np.sin(kR[m])-kR[m]*np.cos(kR[m]))/kR[m]**3
    return float(np.sqrt(np.mean(
        np.real(np.fft.ifftn(dk*W))**2)))

```

Peer Reviews and Revision History

First Round

Reviewer recommendation: **Major Revision.**

The reviewer identified five major and seven minor issues. All have been addressed in this revision:

Major Issue 1 (Promise–delivery gap): The original submission presented no N-body results for $\alpha \neq 0$. This revision presents PM simulation outputs at 64^3 for five values of α : power spectra (Figure 2), σ_8 ratios (Table 2, Figure 3), and density field slices (Figure 4).

Major Issue 2 (Convergence study): A convergence study at 32^3 and 64^3 is now included (Table 1, Figure 1), demonstrating that the σ_8 ratio converges at the $\sim 1.5\%$ level.

Major Issue 3 (Incomplete code): The `cic.interp` function and all referenced functions are now complete in Appendix A. The full simulation code is provided as the supplementary file `run_sims.py`.

Major Issue 4 (Phase transition self-consistency): The limitation of the

sequestration assumption during the phase transition is now explicitly discussed in Remark 5.2. The phase transition results are labelled as exploratory predictions conditional on this assumption.

Major Issue 5 (Statistical rigor): Table 3 and Figure 5 are now explicitly labelled as “conditional on fixed Planck cosmology.”

Minor Issues 6–12: Addressed as follows. (6) The derivation of Eq. (2) from the ansatz is now shown explicitly. (7) Counterarguments sections removed; objections integrated into Discussion (§7). (8) Void fraction parameterization motivated and sensitivity discussed (§5). (9) Press–Schechter vs. Sheth–Tormen comparison added. (10) Code–equation correspondence clarified (Remark 5.1). (11) Density slices from actual simulations replace Zel’dovich visualizations. (12) Complete code provided.

Second Round

Placeholder: to be completed upon receipt of second-round reviews.

Further Clarifications

Placeholder: to be completed as needed.

Extended Formalization

Placeholder: to be completed as needed.